# CHAPTER IV

## NATURAL SELECTION; OR THE SURVIVAL OF THE FITTEST

*Natural Selection—its power compared with man's selection—its power on characters of trifling importance—its power at all ages and on both sexes—Sexual selection—On the generality of intercrosses between individuals of the same species—Circumstances favourable and unfavourable to the results of natural selection, namely, intercrossing, isolation, number of individuals-Slow action-Extinction caused by natural selection —Divergence of character, related to the diversity of inhabitants of any small area, and to naturalisation-Action of natural selection, through divergence of character and extinction, on the descendants from a common parent—Explains the grouping of all* organic *beings-Advance in organisation-Low forms preserved—Convergence* of *character—Indefinite multiplication* of *species—Summary.*

How WILL THE struggle for existence, briefly discussed in the last chapter, act in regard to variation? Can the principle of selection, which we have seen is so potent in the hands of man, apply under nature? I think we shall see that it can act most efficiently. Let the endless number of slight variations and individual differences occurring in our domestic productions, and, in a lesser degree, in those under nature, be borne in mind; as well as the strength of the hereditary tendency. Under domestication, it may be truly said that the whole organisation becomes in some degree plastic. But the variability, which we almost universally meet with in our domestic productions, is not directly produced, as Hooker and Asa Gray have well remarked, by man; he can neither originate varieties, nor prevent their occurrence; he can preserve and accumulate such as do occur. Unintentionally he exposes organic beings to new and changing conditions of life, and variability ensues; but similar changes of conditions might and do occur under nature. Let it also be borne in mind how infinitely complex and close-fitting are the mutual relations of all organic beings to each other and to their physical conditions of life; and consequently what infinitely varied diversities of structure might be of use to each being under changing conditions of life. Can it, then, be thought